Machine Learning II
2026/2027 (CC3043), DCC/FCUP

Project 2026/2027
Submission deadline: November 30th, 2026, 23:59

This work will need to be submitted using the Moodle platform. It will be developed during the practical classes, but it is expected that the students will complement this work using extra-class hours.

1 Objective

The objective of this project is to develop deep learning classifiers that detect heart murmurs from cardiac auscultation recordings (phonocardiograms).

2 Dataset

You will use the **CirCor DigiScope** dataset. The full dataset contains 5272 heart sound recordings from 1568 subjects aged between 0 and 21 years. Recordings were collected at up to four auscultation locations per subject (aortic, pulmonary, tricuspid, and mitral valves) and last between 4.8 and 80.4 seconds.

Only about 60% of the dataset is publicly available: **3163 recordings from 942 patients**. You will train, validate, and test your models using only this public portion.

The dataset description and download instructions are available at:

- Dataset page: `https://physionet.org/content/circor-heart-sound/1.0.3/`
- Reference paper: `https://ieeexplore.ieee.org/abstract/document/9658215`

The dataset provides several labels per patient (demographic information and clinical findings). In this project you will use **only the label `Murmur`**, which can take three values:

- *Present*: a murmur was heard;
- *Absent*: no murmur was heard;
- *Unknown*: the expert could not determine whether a murmur was present (e.g., due to noise or poor recording quality).

This is the same classification task as the murmur detection part of *The George B. Moody PhysioNet Challenge 2022*: `https://physionet.org/content/challenge-2022/1.0.0/`

Important points to keep in mind:

- **The label refers to the patient, not to the individual recording.** A patient typically has several recordings (one per auscultation location), and a murmur may be audible only at some of them (see the

`Murmur locations` field). Your final system must produce **exactly one prediction per patient**. How you combine information from multiple recordings is a design choice you must make and justify (e.g., aggregating recording-level predictions, or feeding all recordings of a patient to the model).

- **The classes are imbalanced**: most patients are labelled *Absent*, and *Unknown* is the smallest class. Take this into account during training and evaluation.

Take your time to explore the data and understand the contents of the dataset before designing your models.

3 Implementation

You must implement **two of the following three** classifiers, at your choice:

- a multilayer perceptron (MLP);
- a convolutional neural network (CNN);
- a recurrent neural network (RNN).

For each chosen classifier, your work must address the following four steps, described in the subsections below:

1. Data pre-processing and preparation
2. Model architecture definition
3. Training strategy
4. Performance evaluation

3.1 Data pre-processing and preparation

In this step you build the input of your neural networks from the raw audio files. Recordings may have different durations, so you will need to make the inputs uniform (e.g., resampling, cropping or splitting into fixed-length windows, padding) and normalize them. You must also decide whether the network receives the raw signal or features extracted from it (e.g., spectrograms, mel-spectrograms, MFCCs). You may use existing libraries (e.g., `librosa`, `torchaudio`, `scipy`) for signal processing and feature extraction.

Note: different architectures may require different input formats. For example, an MLP needs a fixed-size vector, a CNN can take a 1D signal or a 2D time-frequency representation, and an RNN takes a sequence.

3.2 Model architecture definition

MLP. Define the number of layers, the number of neurons in each layer, and the activation function of each layer.

CNN. First decide whether to use 1D or 2D inputs. A 1D CNN is applied directly to (windows of) the audio signal, after downsampling and normalization. A 2D CNN is applied to a time-frequency representation of the sound, such as a spectrogram, a mel-spectrogram, or a sequence of MFCC vectors. Then define the architecture: number and type of layers, filter sizes, number of feature maps, pooling, etc.

RNN. Choose the type of recurrent unit (e.g., vanilla RNN, gated recurrent unit (GRU), long short-term memory (LSTM)) and the type of connections (unidirectional or bidirectional), as well as the number of layers and hidden units.

3.3 Training strategy

Define and justify the details of how your models are trained, including:

- the loss function (and how you handle class imbalance, e.g., class weights or resampling);
- the optimizer;
- the learning hyperparameters (e.g., learning rate, mini-batch size, number of epochs) and how you selected them;
- regularization techniques (e.g., early stopping, weight decay, dropout, data augmentation);
- possibly, the use of transfer learning;
- etc.

3.4 Performance evaluation

Data splitting. You must define your own evaluation protocol on the public data, e.g., a training/validation/test split or cross-validation. The following rules apply:

- **Split by patient, not by recording.** All recordings of the same patient must belong to the same subset; otherwise, your results will be overly optimistic.
- **The test set must not be used to make any decision about your model** (architecture, hyperparameters, pre-processing, early stopping, threshold selection, etc.). Use the validation set (or inner cross-validation folds) for these decisions.
- Consider stratifying the split so that each subset has similar class proportions.

Evaluation metric. The main metric used to evaluate your models is **the same metric used for the murmur task of the PhysioNet Challenge 2022**, the **weighted accuracy**. It is computed **at the patient level** (one prediction per patient) from the confusion matrix in Table 1, where m_{ij} is the number of patients **predicted** as class i whose **true** class is j, with $i, j \in \{P, U, A\}$ (*Present*, *Unknown*, *Absent*):

		True label		
		Present	*Unknown*	*Absent*
Predicted	*Present*	m_{PP}	m_{PU}	m_{PA}
	Unknown	m_{UP}	m_{UU}	m_{UA}
	Absent	m_{AP}	m_{AU}	m_{AA}

Table 1: Patient-level confusion matrix for the murmur task.

$$\mathrm{WA}_{\text{murmur}} = \frac{5\,m_{PP} + 3\,m_{UU} + m_{AA}}{5\,(m_{PP} + m_{UP} + m_{AP}) + 3\,(m_{PU} + m_{UU} + m_{AU}) + (m_{PA} + m_{UA} + m_{AA})}.$$

In words, it is the accuracy obtained when each *Present* patient counts 5 times, each *Unknown* patient counts 3 times, and each *Absent* patient counts once. These weights reflect the clinical cost of errors: missing a murmur (or a patient that needs further examination) is worse than a false alarm. The metric ranges from 0 to 1 (higher is better). Note that a trivial classifier that always predicts *Absent* achieves a high standard accuracy but a low weighted accuracy.

The official implementation of the metric is available in the Challenge's evaluation code (`https://github.com/physionetchallenges/evaluation-2022`). You may use it directly or reimplement it, but your result must match the definition above.

What to report. For each classifier, report on your test data (or across cross-validation folds, with mean and standard deviation):

- the weighted accuracy $\mathrm{WA}_{\mathrm{murmur}}$ (**mandatory**);
- the patient-level confusion matrix (**mandatory**);
- any other metrics you find useful (e.g., per-class recall, F1-score), clearly identified as additional.

The same weighted accuracy will be used to rank the groups in the bonus competition (Section 4).

4 Bonus

The project is graded out of 8 points. The bonus is worth up to 1 additional point, but the maximum project grade remains 8.

The bonus is a competition on hidden data, i.e., patient recordings that are not publicly available and whose labels you do not know. These recordings will be shared later during the semester. You will submit the predictions of your best model for each patient in this set, and groups will be ranked according to the weighted accuracy $\mathrm{WA}_{\mathrm{murmur}}$ defined in Section 3.4:

- groups ranked 1st to 5th receive 1 point;
- groups ranked 6th to 10th receive 0.5 points.

The predictions must be submitted as a .csv file with one row per patient and two columns: the patient ID and the predicted label (`Present`, `Unknown`, or `Absent`). Example:

```
patient_id,murmur
12345,Absent
12346,Present
```

5 Submission

Submit the following material on Moodle by **November 30, 2026, at 23:59**:

1. **Code:** a notebook or a set of files that allows to train your models and reproduce the reported results. **Do not include the dataset files**; assume the dataset is available locally. Include brief instructions on how to run the code.

2. **Presentation slides** (PDF) for a **10-minute presentation**. Additional material that does not fit in that time may be included in an appendix. The slides must contain:
    - a brief description of the deep learning approaches you chose (two out of MLP, CNN, RNN);
    - **the implementation description (most important part):** motivate your choices for pre-processing and data preparation, model design, and training strategy, and explain how you obtained one prediction per patient. Show intermediate results, how you interpreted them, and what you changed to improve them;
    - **results:** the weighted accuracy and confusion matrix of each classifier, obtained with the evaluation protocol of Section 3.4;
    - **discussion and conclusions:** comments on the results, comparison between classifiers, limitations, and future work.
3. **Predictions on hidden data:** a .csv file in the format described in Section 4.
4. **Self-assessment:** the completed form describing the contribution of each group member.

6 Rules

- Projects are developed in groups of **3 students**. Smaller groups must be justified and approved before starting the work.
- During the practical classes of November 5, 2026, each group will undergo a brief check point verification of the status of their work. This will not require a formal presentation, but you will need to be able to explain what you have been doing up to that point. The check point will be evaluated and will account for 10% of the project grade.
- Late submissions will be penalized and may not be accepted.
- All projects will be presented on **December 3 and December 10, 2026**. All group members must attend. **A student who is not present at the presentation receives a grade of zero for the project, and therefore fails it.**
- Each member must describe their own contribution and must also know in detail what the other members did. **Students who cannot explain in detail what the solution does and why will be penalized.**